```python
import pandas as pd
```

```python
pd.__version__
```

```
'2.3.3'
```

```python
emp = pd.read_excel(r'E:\PYTHON DATA STRUCTURES\Rawdata.xlsx')
emp
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience#$	34 years	Mumbai	5^00#0	2+
1	Teddy^	Testing	45' yr	Bangalore	10%%000	<3
2	Uma#r	Dataanalyst^^#	NaN	NaN	1$5%000	4> yrs
3	Jane	Ana^^lytics	NaN	Hyderbad	2000^0	NaN
4	Uttam*	Statistics	67-yr	NaN	30000-	5+ year
5	Kim	NLP	55yr	Delhi	6000^$0	10+

data cleaning

```python
emp['Name']
```

```
0      Mike
1    Teddy^
2     Uma#r
3      Jane
4    Uttam*
5       Kim
Name: Name, dtype: object
```

```python
emp['Name'] = emp['Name'].str.replace(r'\W', '', regex = True)
emp['Name']
```

```
0     Mike
1    Teddy
2     Umar
3     Jane
4    Uttam
5      Kim
Name: Name, dtype: object
```

```python
emp.columns
```

```
Index(['Name', 'Domain', 'Age', 'Location', 'Salary', 'Exp'],
dtype='object')
```

```python
emp.shape
```

```
(6, 6)
```

```python
emp.Location
```

```
0
1
2    NaN
3
4    NaN
5
Name: Location, dtype: object
```

```
emp.head()
```

```
    Name       Domain  Age Location Salary  Exp
0   Mike  Datascience   34            5000    2
1  Teddy      Testing   45           10000    3
2   Umar  Dataanalyst  NaN      NaN  15000    4
3   Jane    Analytics  NaN           20000  NaN
4  Uttam   Statistics   67      NaN  30000    5
```

```
emp.head(1)
```

```
   Name         Domain       Age Location  Salary Exp
0  Mike  Datascience#$  34 years   Mumbai  5^00#0  2+
```

```
emp.tail()
```

```
    Name       Domain  Age Location Salary  Exp
1  Teddy      Testing   45           10000    3
2   Umar  Dataanalyst  NaN      NaN  15000    4
3   Jane    Analytics  NaN           20000  NaN
4  Uttam   Statistics   67      NaN  30000    5
5    Kim          NLP   55           60000   10
```

```
emp.tail(3)
```

```
    Name      Domain  Age Location Salary  Exp
3   Jane   Analytics  NaN           20000  NaN
4  Uttam  Statistics   67      NaN  30000    5
5    Kim         NLP   55           60000   10
```

```
emp.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Name      6 non-null      object
 1   Domain    6 non-null      object
 2   Age       4 non-null      object
 3   Location  4 non-null      object
 4   Salary    6 non-null      object
 5   Exp       5 non-null      object
```

```
dtypes: object(6)
memory usage: 420.0+ bytes

emp

    Name       Domain  Age Location Salary  Exp
0   Mike  Datascience   34            5000    2
1  Teddy      Testing   45           10000    3
2   Umar  Dataanalyst  NaN      NaN  15000    4
3   Jane    Analytics  NaN           20000  NaN
4  Uttam   Statistics   67      NaN  30000    5
5    Kim          NLP   55           60000   10

emp['Domain']

0     Datascience#$
1           Testing
2    Dataanalyst^^#
3       Ana^^lytics
4        Statistics
5               NLP
Name: Domain, dtype: object

emp.isnull()

    Name  Domain    Age  Location  Salary    Exp
0  False   False  False     False   False  False
1  False   False  False     False   False  False
2  False   False   True      True   False  False
3  False   False   True     False   False   True
4  False   False  False      True   False  False
5  False   False  False     False   False  False

emp.isnull().sum()

Name        0
Domain      0
Age         2
Location    2
Salary      0
Exp         1
dtype: int64

emp['Location'] = emp['Location'].str.replace(r'[^0-9]', '', regex =
True)
emp

    Name       Domain  Age Location Salary  Exp
0   Mike  Datascience   34            5000    2
1  Teddy      Testing   45           10000    3
2   Umar  Dataanalyst  NaN      NaN  15000    4
```

```
3   Jane    Analytics  NaN           20000  NaN
4  Uttam   Statistics   67      NaN  30000    5
5    Kim          NLP   55           60000   10
```

```
emp['Location'] = emp['Location'].str.strip()
emp['Location']
```

```
0
1
2    NaN
3
4    NaN
5
Name: Location, dtype: object
```

```
emp['Location'] = emp['Location'].str.replace(r'\W','', regex=True)
emp
```

```
    Name       Domain  Age Location Salary  Exp
0   Mike  Datascience   34            5000    2
1  Teddy      Testing   45           10000    3
2   Umar  Dataanalyst  NaN      NaN  15000    4
3   Jane    Analytics  NaN           20000  NaN
4  Uttam   Statistics   67      NaN  30000    5
5    Kim          NLP   55           60000   10
```

```
emp['Age'] = emp['Age'].str.replace(r'\W', '', regex = True)
emp['Age']
```

```
0    34years
1       45yr
2        NaN
3        NaN
4       67yr
5       55yr
Name: Age, dtype: object
```

```
emp['Age'] = emp['Age'].str.extract('(\\d+)') # r(r'(\\d+)')
emp['Age']
```

```
0     34
1     45
2    NaN
3    NaN
4     67
5     55
Name: Age, dtype: object
```

```
emp
```

```
    Name       Domain  Age Location   Salary      Exp
0   Mike  Datascience   34            5^00#0       2+
1  Teddy      Testing   45           10%%000       <3
2   Umar  Dataanalyst  NaN      NaN  1$5%000   4> yrs
3   Jane    Analytics  NaN            2000^0      NaN
4  Uttam   Statistics   67      NaN   30000-  5+ year
5    Kim          NLP   55           6000^$0      10+
```

```python
emp['Salary'] = emp['Salary'].str.replace(r'\W', '', regex = True)
emp['Salary']
```

```
0     5000
1    10000
2    15000
3    20000
4    30000
5    60000
Name: Salary, dtype: object
```

```python
emp
```

```
    Name       Domain  Age Location Salary      Exp
0   Mike  Datascience   34            5000       2+
1  Teddy      Testing   45           10000       <3
2   Umar  Dataanalyst  NaN      NaN  15000   4> yrs
3   Jane    Analytics  NaN           20000      NaN
4  Uttam   Statistics   67      NaN  30000  5+ year
5    Kim          NLP   55           60000      10+
```

```python
emp['Exp'] = emp['Exp'].str.extract('(\\d+)')
emp['Exp']
```

```
0      2
1      3
2      4
3    NaN
4      5
5     10
Name: Exp, dtype: object
```

```python
emp
```

```
    Name       Domain  Age Location Salary  Exp
0   Mike  Datascience   34            5000    2
1  Teddy      Testing   45           10000    3
2   Umar  Dataanalyst  NaN      NaN  15000    4
3   Jane    Analytics  NaN           20000  NaN
4  Uttam   Statistics   67      NaN  30000    5
5    Kim          NLP   55           60000   10
```

```python
clean_data = emp.copy()
```

```
clean_data
```

```
    Name       Domain  Age Location Salary  Exp
0   Mike  Datascience   34            5000    2
1  Teddy      Testing   45           10000    3
2   Umar  Dataanalyst  NaN      NaN  15000    4
3   Jane    Analytics  NaN           20000  NaN
4  Uttam   Statistics   67      NaN  30000    5
5    Kim          NLP   55           60000   10
```

```
clean_data['Age']
```

```
0     34
1     45
2    NaN
3    NaN
4     67
5     55
Name: Age, dtype: object
```

```
import numpy as np
```

```
clean_data['Age'] =
clean_data['Age'].fillna(np.mean(pd.to_numeric(clean_data['Age'])))
clean_data['Age']
```

```
0       34
1       45
2    50.25
3    50.25
4       67
5       55
Name: Age, dtype: object
```

```
clean_data['Exp']
```

```
0      2
1      3
2      4
3    NaN
4      5
5     10
Name: Exp, dtype: object
```

```
clean_data['Exp'] =
clean_data['Exp'].fillna(np.mean(pd.to_numeric(clean_data['Exp'])))
clean_data['Exp']
```

```
0      2
1      3
2      4
```

```
3    4.8
4      5
5     10
Name: Exp, dtype: object
```

```
clean_data
```

```
    Name       Domain    Age Location Salary  Exp
0   Mike  Datascience     34            5000    2
1  Teddy      Testing     45           10000    3
2   Umar  Dataanalyst  50.25      NaN  15000    4
3   Jane    Analytics  50.25           20000  4.8
4  Uttam   Statistics     67      NaN  30000    5
5    Kim          NLP     55           60000   10
```

```
clean_data['Location'].isnull().sum()
clean_data['Location']
```

```
0
1
2    NaN
3
4    NaN
5
Name: Location, dtype: object
```

```
clean_data['Location'].isnull().sum()
```

```
np.int64(2)
```

```
clean_data['Location']
```

```
0
1
2    NaN
3
4    NaN
5
Name: Location, dtype: object
```

```
clean_data['Location'] =
clean_data['Location'].fillna(clean_data['Location'].mode()[0])
clean_data['Location']
```

```
0
1
2
3
4
5
Name: Location, dtype: object
```

```
clean_data
```

```
    Name       Domain    Age Location Salary  Exp
0   Mike  Datascience     34           5000    2
1  Teddy      Testing     45          10000    3
2   Umar  Dataanalyst  50.25          15000    4
3   Jane    Analytics  50.25          20000  4.8
4  Uttam   Statistics     67          30000    5
5    Kim          NLP     55          60000   10
```

```
clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype 
---  ------    --------------  ----- 
 0   Name      6 non-null      object
 1   Domain    6 non-null      object
 2   Age       6 non-null      object
 3   Location  6 non-null      object
 4   Salary    6 non-null      object
 5   Exp       6 non-null      object
dtypes: object(6)
memory usage: 420.0+ bytes
```

```
clean_data['Age'] = clean_data['Age'].astype(int)
clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype 
---  ------    --------------  ----- 
 0   Name      6 non-null      object
 1   Domain    6 non-null      object
 2   Age       6 non-null      int64 
 3   Location  6 non-null      object
 4   Salary    6 non-null      object
 5   Exp       6 non-null      object
dtypes: int64(1), object(5)
memory usage: 420.0+ bytes
```

```
clean_data['Salary'] = clean_data['Salary'].astype(int)
clean_data['Exp'] = clean_data['Exp'].astype(int)

clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
```

```
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Name      6 non-null      object
 1   Domain    6 non-null      object
 2   Age       6 non-null      int64
 3   Location  6 non-null      object
 4   Salary    6 non-null      int64
 5   Exp       6 non-null      int64
dtypes: int64(3), object(3)
memory usage: 420.0+ bytes
```

```
clean_data['Name'] = clean_data['Name'].astype('category')
clean_data['Domain'] = clean_data['Domain'].astype('category')
clean_data['Location'] = clean_data['Location'].astype('category')
```

```
clean_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Name      6 non-null      category
 1   Domain    6 non-null      category
 2   Age       6 non-null      int64
 3   Location  6 non-null      category
 4   Salary    6 non-null      int64
 5   Exp       6 non-null      int64
dtypes: category(3), int64(3)
memory usage: 850.0 bytes
```

```
clean_data
```

```
    Name       Domain  Age Location  Salary  Exp
0   Mike  Datascience   34             5000    2
1  Teddy      Testing   45            10000    3
2   Umar  Dataanalyst   50            15000    4
3   Jane    Analytics   50            20000    4
4  Uttam   Statistics   67            30000    5
5    Kim          NLP   55            60000   10
```

```
clean_data.to_csv('clean_data.csv')
```

```
import os
os.getcwd()
```

```
'C:\\Users\\ABHINAV'
```

```
clean_data
```

	Name	Domain	Age	Location	Salary	Exp
0	Mike	Datascience	34		5000	2
1	Teddy	Testing	45		10000	3
2	Umar	Dataanalyst	50		15000	4
3	Jane	Analytics	50		20000	4
4	Uttam	Statistics	67		30000	5
5	Kim	NLP	55		60000	10